

Samuele Doria

✉ samueledoria@gmail.com | 🏠 samudoria.github.io | 💻 github.com/samudoria | 🔗 linkedin.com/in/samueledoria | 🎓 Google Scholar Profile

Personal Profile

Security Engineer and Researcher specializing in Software and Mobile Security. Focused on engineering practical solutions for vulnerability detection, leveraging advanced static analysis and hybrid techniques. Author of multiple high-impact security tools developed in collaboration with industry leaders like Google. Active participant in the competitive security community with a strong foundation in Reverse Engineering and Binary Exploitation.

Work Experience

Teaching Assistant

University of Padua

Padua, Italy

Feb 2026 - July 2026

- Teaching Assistant for the “Blockchain and Smart Contract” course.

Visiting PhD Researcher

EURECOM

Biot, France

Sep 2025 - Mar 2026

- Architected an automated binary analysis agent using the Model Context Protocol (MCP) to orchestrate interaction between Large Language Models (LLMs) and reverse engineering tools.
- Conducted a comparative benchmark between proprietary models, open-source alternatives, and human experts to evaluate automated reverse engineering capabilities.

Mobile Security Researcher

RCH Group Spa

Treviso, Italy

Sep 2023 - Mar 2025

- Performed security assessments of Android payment and self-checkout applications using static and dynamic analysis tools, including MobSF, SPECK, and Frida.
- Conducted research and testing on Android application protection mechanisms, including obfuscation, APK tampering, and Google Play Integrity API / SafetyNet bypass techniques.
- Evaluated secure data storage, inter-app communication, and OS integrity for Android-based payment and SoftPOS systems.

Teaching Assistant

University of Padua

Padua, Italy

Sep 2024 - Feb 2025

- Teaching Assistant for the “Mobile Security” course.

Research Intern at HexHive

École Polytechnique Fédérale de Lausanne (EPFL)

Lausanne, Switzerland

May 2023 - Aug 2023

- Developed a static analysis tool to create fuzzing harnesses for Android native libraries.

Peer-Tutor for CyberChallenge.it

University of Calabria

Cosenza, Italy

Feb 2021 - July 2021

- Peer-tutoring for the [CyberChallenge.it](#) initiative's participants. I guided them in their first experiences playing CTFs, more specifically in challenges involving Reverse Engineering and Binary Exploitation.

Education

PhD Student

University of Padua

Padua, Italy

Nov 2023 - Current

- Researching Software Security, with a focus on Android Security and AI for Security.
- Anticipated graduation date: November 2026.

MSc in CyberSecurity

University of Padua

Padua, Italy

Sept 2021 - Sept 2023

- Graduation score: 110/110 cum laude
- **Thesis:** [Control-Flow Graph Based Path Reconstruction in Android Applications](#)

BSc in Computer Engineering

University of Calabria

Cosenza, Italy

Oct 2018 - Sept 2021

- Graduation score: 107/110
- **Thesis:** Binary Exploitation on x86-64 and ARM

CyberChallenge.it Participant

University of Calabria

Cosenza, Italy

Feb 2020 - June 2020

- Selected among 20 participants to attend lectures and trainings on CyberSecurity and CTF challenges.

Talks & Projects

- Project: “SPECK: From Android Textual Guidelines To Automatic Exploitation”**

Collaboration with Google

Funded by the [Google Research Scholar Program](#) (“Security” category)

November 2022 - Current

- Engineered **SPECK**, a rule-based static analyzer compliant with Google’s official security guidelines to automatically detect vulnerabilities in Android applications.
 - Developed **GAPS**, a hybrid analysis tool focusing on code reachability to validate findings and reduce false positives.
- Project: “Decompiling the Synergy: Human-LLM Teaming in Reverse Engineering”**

San Diego, USA

Published at NDSS 2026

February 2026

- Architected **REaLLM**, a cross-platform AI agent (IDA Pro, Ghidra) that automates artifact recovery for reverse engineers.
 - Orchestrated a large-scale empirical study (n=40), demonstrating that AI assistance enables novices to match expert performance in comprehension tasks.
 - Quantified the risks of LLM hallucinations, proving that AI-driven vulnerability detection currently yields a negative impact on audit accuracy.
- Speaker at DEF CON 33 at the Mobile Hacking Community**

Las Vegas, USA

DEF CON 33

August 2025

- Selected to present original research on Android virtualization-based malware.
 - Demonstrated practical detection techniques and defense methodologies to industry professionals and security researchers.
- Project: “Hercules Droidot and the Murder on the JNI Express”**

Seattle, USA

Published at USENIX Security Symposium

August 2025

- Co-developed **Poirot**: a fuzzing framework to identify memory corruption vulnerabilities at the Java Native Interface (JNI) boundary.
 - Automated the detection of unsafe JNI patterns in large-scale Android applications.
- Project: “VirtualPatch: Distributing Android Security Patches through Virtualization”**

Published at Computers & Security Journal

August 2025

- Designed a novel mechanism to hot-patch Android vulnerabilities using virtualization layers, bypassing system-level updates.
 - Developed 10 exploits and patches following real-world CVEs to prove **VirtualPatch** effectiveness.
- Project: “Matrioska: A User-Centric Defense Against Repackaging Malware”**

Honolulu, USA

Published at Annual Computer Security Applications Conference (ACSAC)

December 2023

- Matrioska** is a defensive framework to detect virtualization-based repackaging malware on Android.
 - Implemented user-centric sandbox to warn users of potential malware.

Honors & Awards

2024	Winner of 5th best Master thesis on CyberSecurity, <i>Clusit</i>	Italy
2023	“Mille e una lode” Award , Merit-based scholarship awarded to the top 3% students of each degree.	Padova, Italy
2021	Scholarship , Awarded for merit during my Master’s studies	Padova, Italy
2018	Scholarship , Awarded for merit during my Bachelor studies	Cosenza, Italy

Skills

Security & Analysis	Binary Exploitation (ARM/x64), Reverse Engineering, Fuzzing (AFL++), Static & Dynamic Analysis, Ghidra, IDA Pro, Frida.
Programming	Python, C/C++, Java, SQL, Assembly (ARM/x64), Bash/Zsh.
Tools & Platforms	Linux, Git, Docker, LaTeX, CI/CD Pipelines.

Languages

Italian	Native proficiency
English	C2 level: proficient user